

Customer-trends-data-analysis

```
In [ ]: import os

print(os.path.exists(r'D:\customer_shopping_behavior.csv'))
```

```
In [2]: import pandas as pd

df = pd.read_csv(r'D:\customer_shopping_behavior.csv')
```

```
In [3]: import pandas as pd

pd.set_option('display.max_columns', None)
pd.set_option('display.width', None)

df.head()
```

Out[3]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location
0	1	55	Male	Blouse	Clothing	53	Kentucky
1	2	19	Male	Sweater	Clothing	64	Maine
2	3	50	Male	Jeans	Clothing	73	Massachusetts
3	4	21	Male	Sandals	Footwear	90	Rhode Island
4	5	45	Male	Blouse	Clothing	49	Oregon

```
In [7]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Customer ID                          3900 non-null   int64
1   Age                                  3900 non-null   int64
2   Gender                              3900 non-null   object
3   Item Purchased                       3900 non-null   object
4   Category                            3900 non-null   object
5   Purchase Amount (USD)                3900 non-null   int64
6   Location                             3900 non-null   object
7   Size                                 3900 non-null   object
8   Color                                3900 non-null   object
9   Season                               3900 non-null   object
10  Review Rating                        3863 non-null   float64
11  Subscription Status                  3900 non-null   object
12  Shipping Type                       3900 non-null   object
13  Discount Applied                    3900 non-null   object
14  Promo Code Used                     3900 non-null   object
15  Previous Purchases                   3900 non-null   int64
16  Payment Method                      3900 non-null   object
17  Frequency of Purchases               3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

In [8]:

Summary statistics using .describe()
df.describe(include='all')

Out[8]:

	Customer ID	Age	Gender	Item Purchased	Category	
count	3900.000000	3900.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN
top	NaN	NaN	Male	Blouse	Clothing	NaN
freq	NaN	NaN	2652	171	1737	NaN
mean	1950.500000	44.068462	NaN	NaN	NaN	59.0
std	1125.977353	15.207589	NaN	NaN	NaN	23.0
min	1.000000	18.000000	NaN	NaN	NaN	20.0
25%	975.750000	31.000000	NaN	NaN	NaN	39.0
50%	1950.500000	44.000000	NaN	NaN	NaN	60.0
75%	2925.250000	57.000000	NaN	NaN	NaN	81.0
max	3900.000000	70.000000	NaN	NaN	NaN	100.0

```
In [9]: # Checking if missing data or null values are present in the dataset

df.isnull().sum()
```

```
Out[9]: Customer ID          0
        Age                  0
        Gender                0
        Item Purchased        0
        Category              0
        Purchase Amount (USD)  0
        Location              0
        Size                  0
        Color                 0
        Season                0
        Review Rating         37
        Subscription Status    0
        Shipping Type          0
        Discount Applied       0
        Promo Code Used        0
        Previous Purchases     0
        Payment Method         0
        Frequency of Purchases  0
        dtype: int64
```

```
In [10]: # Imputing missing values in Review Rating column with the median rating of
         the product category

df['Review Rating'] = df.groupby('Category')['Review
Rating'].transform(lambda x: x.fillna(x.median()))
```

```
In [11]: df.isnull().sum()
```

```
Out[11]: Customer ID      0
         Age              0
         Gender           0
         Item Purchased   0
         Category         0
         Purchase Amount (USD) 0
         Location         0
         Size             0
         Color            0
         Season           0
         Review Rating    0
         Subscription Status 0
         Shipping Type    0
         Discount Applied 0
         Promo Code Used  0
         Previous Purchases 0
         Payment Method   0
         Frequency of Purchases 0
         dtype: int64
```

```
In [12]: # Renaming columns according to snake casing for better readability and
         documentation
```

```
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ','_')
df = df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})
```

```
In [13]: df.columns
```

```
Out[13]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
               'purchase_amount', 'location', 'size', 'color', 'season',
               'review_rating', 'subscription_status', 'shipping_type',
               'discount_applied', 'promo_code_used', 'previous_purchases',
               'payment_method', 'frequency_of_purchases'],
              dtype='object')
```

```
In [15]: #create a new column age_group
labels = ['Young Adult', 'Adult', 'Middle-aged', 'Senior']
df['age_group'] = pd.qcut(df['age'], q=4, labels = labels)
```

```
In [17]: # create new column purchase_frequency_days

frequency_mapping = {
    'Fortnightly': 14,
    'Weekly': 7,
    'Monthly': 30,
    'Quarterly': 90,
    'Bi-Weekly': 14,
    'Annually': 365,
    'Every 3 Months': 90
}

df['purchase_frequency_days'] =
df['frequency_of_purchases'].map(frequency_mapping)
```

```
In [18]: df[['purchase_frequency_days', 'frequency_of_purchases']].head(10)
```

Out[18]:

	purchase_frequency_days	frequency_of_purchases
0	14	Fortnightly
1	14	Fortnightly
2	7	Weekly
3	7	Weekly
4	365	Annually
5	7	Weekly
6	90	Quarterly
7	7	Weekly
8	365	Annually
9	90	Quarterly

```
In [19]: df[['discount_applied', 'promo_code_used']].head(10)
```

Out[19]:

	discount_applied	promo_code_used
0	Yes	Yes
1	Yes	Yes
2	Yes	Yes
3	Yes	Yes
4	Yes	Yes
5	Yes	Yes
6	Yes	Yes
7	Yes	Yes
8	Yes	Yes
9	Yes	Yes

```
In [20]: (df['discount_applied'] == df['promo_code_used']).all()
```

Out[20]: np.True_

```
In [21]: # Dropping promo code used column

df = df.drop('promo_code_used', axis=1)
```

```
In [22]: df.columns
```

Out[22]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
'purchase_amount', 'location', 'size', 'color', 'season',
'review_rating', 'subscription_status', 'shipping_type',
'discount_applied', 'previous_purchases', 'payment_method',
'frequency_of_purchases', 'age_group',
'purchase_frequency_days'],
dtype='object')

In [24]: `df.head(5)`

Out[24]:

	customer_id	age	gender	item_purchased	category	purchase_am
0	1	55	Male	Blouse	Clothing	53
1	2	19	Male	Sweater	Clothing	64
2	3	50	Male	Jeans	Clothing	73
3	4	21	Male	Sandals	Footwear	90
4	5	45	Male	Blouse	Clothing	49

Connecting Python script to PostgreSQL

In [23]: `!pip install psycopg2-binary sqlalchemy`

```

Defaulting to user installation because normal site-packages is not
writeable
Collecting psycopg2-binary
  Downloading psycopg2_binary-2.9.12-cp313-cp313-
win_amd64.whl.metadata (5.1 kB)
Requirement already satisfied: sqlalchemy in g:\anaconda\lib\site-
packages (2.0.43)
Requirement already satisfied: greenlet>=1 in g:\anaconda\lib\site-
packages (from sqlalchemy) (3.2.4)
Requirement already satisfied: typing-extensions>=4.6.0 in
g:\anaconda\lib\site-packages (from sqlalchemy) (4.15.0)
Downloading psycopg2_binary-2.9.12-cp313-cp313-win_amd64.whl (2.8 MB)
----- 0.0/2.8 MB ? eta -:-:--
----- 0.0/2.8 MB ? eta -:-:--
----- 1.0/2.8 MB 6.7 MB/s eta
0:00:01
----- 1.0/2.8 MB 6.7 MB/s eta
0:00:01
----- 2.8/2.8 MB 4.3 MB/s
0:00:00
Installing collected packages: psycopg2-binary
Successfully installed psycopg2-binary-2.9.12

```



```
In [5]: !pip install pymysql sqlalchemy
```

```
Defaulting to user installation because normal site-packages is not writeable
```

```
Collecting pymysql
```

```
  Downloading pymysql-1.2.0-py3-none-any.whl.metadata (4.3 kB)
```

```
Requirement already satisfied: sqlalchemy in g:\anaconda\lib\site-packages (2.0.43)
```

```
Requirement already satisfied: greenlet>=1 in g:\anaconda\lib\site-packages (from sqlalchemy) (3.2.4)
```

```
Requirement already satisfied: typing-extensions>=4.6.0 in g:\anaconda\lib\site-packages (from sqlalchemy) (4.15.0)
```

```
Downloading pymysql-1.2.0-py3-none-any.whl (45 kB)
```

```
Installing collected packages: pymysql
```

```
Successfully installed pymysql-1.2.0
```

```
In [16]: import pandas as pd
from sqlalchemy import create_engine

# PostgreSQL connection
username = "postgres"      # PostgreSQL login username
password = "1996"          # Your PostgreSQL password
host = "localhost"
port = "5432"
database = "customer_behavior"

engine = create_engine(
    f"postgresql+psycopg2://{username}:{password}@{host}:{port}/{database}"
)

# Upload DataFrame to PostgreSQL
df.to_sql("customer", engine, if_exists="replace", index=False)

print("Data uploaded successfully!")

# Read data
df_pg = pd.read_sql("SELECT * FROM customer LIMIT 5;", engine)
print(df_pg)
```

Data uploaded successfully!

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)
0	1	55	Male	Blouse	Clothing	53
1	2	19	Male	Sweater	Clothing	64
2	3	50	Male	Jeans	Clothing	73
3	4	21	Male	Sandals	Footwear	90
4	5	45	Male	Blouse	Clothing	49

	Location	Size	Color	Season	Review Rating	Subscription Status
0	Kentucky	L	Gray	Winter	3.1	Yes
1	Maine	L	Maroon	Winter	3.1	Yes
2	Massachusetts	S	Maroon	Spring	3.1	Yes
3	Rhode Island	M	Maroon	Spring	3.5	Yes
4	Oregon	M	Turquoise	Spring	2.7	Yes

	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases
0	Express	Yes	Yes	14

1	Express	Yes	Yes	2
2	Free Shipping	Yes	Yes	23
3	Next Day Air	Yes	Yes	49
4	Free Shipping	Yes	Yes	31

	Payment Method	Frequency of Purchases
0	Venmo	Fortnightly
1	Cash	Fortnightly
2	Credit Card	Weekly
3	PayPal	Weekly
4	PayPal	Annually

In [18]: `df.columns`

Out[18]: Index(['Customer ID', 'Age', 'Gender', 'Item Purchased', 'Category', 'Purchase Amount (USD)', 'Location', 'Size', 'Color', 'Season', 'Review Rating', 'Subscription Status', 'Shipping Type', 'Discount Applied', 'Promo Code Used', 'Previous Purchases', 'Payment Method', 'Frequency of Purchases'], dtype='object')

In [19]: `df.columns = df.columns.str.lower().str.replace(" ", "_")`
`df.rename(`
 `columns={"purchase_amount_(usd)": "purchase_amount"},`
 `inplace=True`
`)`

In [20]: `df.columns`

Out[20]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category', 'purchase_amount', 'location', 'size', 'color', 'season', 'review_rating', 'subscription_status', 'shipping_type', 'discount_applied', 'promo_code_used', 'previous_purchases', 'payment_method', 'frequency_of_purchases'], dtype='object')

In [21]: `df.drop(columns=["promo_code_used"], inplace=True)`

In [22]: `df.columns`

Out[22]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category', 'purchase_amount', 'location', 'size', 'color', 'season', 'review_rating', 'subscription_status', 'shipping_type', 'discount_applied', 'previous_purchases', 'payment_method', 'frequency_of_purchases'], dtype='object')

```

In [23]: import pandas as pd
         from sqlalchemy import create_engine

         # PostgreSQL connection details
         username = "postgres"
         password = "1996"
         host = "localhost"
         port = "5432"
         database = "customer_behavior"

         # Create connection
         engine = create_engine(
             f"postgresql+psycopg2://{username}:{password}@{host}:{port}/{database}"
         )

         # Upload cleaned DataFrame to PostgreSQL
         df.to_sql(
             name="customer",
             con=engine,
             if_exists="replace",  # Replaces the old table
             index=False
         )

         print("✅ Data uploaded successfully!")

         # Verify upload
         df_pg = pd.read_sql(
             "SELECT * FROM customer LIMIT 5;",
             engine
         )

         print(df_pg)

```

```

✅ Data uploaded successfully!
  customer_id  age gender item_purchased  category  purchase_amount
\
0           1   55  Male         Blouse  Clothing             53
1           2   19  Male         Sweater  Clothing             64
2           3   50  Male          Jeans  Clothing             73
3           4   21  Male        Sandals  Footwear             90
4           5   45  Male         Blouse  Clothing             49

      location size      color season  review_rating
subscription_status \
0      Kentucky  L      Gray  Winter             3.1
Yes
1           Maine  L    Maroon  Winter             3.1
Yes
2  Massachusetts  S    Maroon  Spring             3.1
Yes
3    Rhode Island  M    Maroon  Spring             3.5
Yes
4           Oregon  M  Turquoise  Spring             2.7

```

Yes

	shipping_type	discount_applied	previous_purchases	payment_method
\				
0	Express	Yes	14	Venmo
1	Express	Yes	2	Cash
2	Free Shipping	Yes	23	Credit Card
3	Next Day Air	Yes	49	PayPal
4	Free Shipping	Yes	31	PayPal

	frequency_of_purchases
0	Fortnightly
1	Fortnightly
2	Weekly
3	Weekly
4	Annually

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